[2nd Semester] [2025 - 2026]

[Lab 5]
Instructor
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[March25, 2026]
Email
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Hours
[2 Hours]

Lab 5 – Login System Using Cookies (Minimal API)

Learning Objectives

By the end of this lab, students will:

- Understand how login works in backend systems
- Use POST to send login data
- Learn how cookies store user information
- Read cookies from requests
- Learn how to use the Result
- Build a simple login + profile system

Scenario

We want to build a simple system:

1. User enters username and password
2. If correct → system stores username in a cookie
3. User visits profile page → system reads cookie and greets user

Project Structure

```
LoginApp
|
|-- Program.cs
|
|-- wwwroot
    |-- login.html
    |-- profile.html
```

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1. Create ASP.NET Core Empty Project
2. Add wwwroot folder
3. Add HTML files
4. Replace Program.cs with the code bellow
5. Run project (F5)
6. Open /login.html

Backend — Program.cs

```
// Step 1: Setup Application
// Create the application builder (initial setup of ASP.NET Core app)
var builder = WebApplication.CreateBuilder(args);
// Build the application (prepare it to handle requests)
var app = builder.Build();
// Enable default files (like index.html)
app.UseDefaultFiles();
// Enable serving static files (HTML, CSS, JS from wwwroot)
app.UseStaticFiles();
```

```
//Step 2: Login Endpoint (POST)
// Create a POST endpoint at /login
app.MapPost("/login", (HttpContext context, LoginRequest request) =>
{   // Check if username and password are correct
if (request.Username == "admin" && request.Password == "123")
{  // Store the username inside a cookie in the user's browser
context.Response.Cookies.Append("username", request.Username);
// Return success response as JSON
return Results.Ok(new
{  Status = "Success", // indicates operation succeeded
   Message = "Login successful" // message to user
});
}
// If login fails, return error response
return Results.BadRequest(new
{   Status = "Error", // indicates failure
    Message = "Invalid username or password" // error message
});
});
```

```
//Step 3: Profile Endpoint (GET)
```

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```
// Create a GET endpoint at /profile
app.MapGet("/profile", (HttpContext context) =>
{
// Read the cookie named "username" from the request
var username = context.Request.Cookies["username"];

// Check if cookie does not exist or is empty
if (string.IsNullOrEmpty(username))
{
// User is not logged in
return Results.BadRequest(new
{
Status = "Error", // error state
Message = "User not logged in" // message explaining issue
});
}
// If cookie exists, return welcome message
return Results.Ok(new
{
Status = "Success", // success state
Message = $"Welcome {username}" // personalized greeting
});
});
```

```
// This class represents the structure of the login request body
public class LoginRequest
{
// Property to store username sent from client
public string Username { get; set; } = "";

// Property to store password sent from client
public string Password { get; set; } = "";
}
```

```
// Start the web server and listen for incoming requests
app.Run();
```

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Frontend login.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Login</title>
</head>
<body>

<!-- Page title -->
<h2>Login</h2>

<!-- Input field for username -->
<input type="text" id="username" placeholder="Username"><br><br>

<!-- Input field for password -->
<input type="password" id="password" placeholder="Password"><br><br>

<!-- Button that triggers login function -->
<button onclick="login()">Login</button>

<!-- Area to display result -->
<h3 id="result"></h3>

<script>

// Function that runs when user clicks login button
function login() {

  // Send POST request to backend
  fetch("/login", {

    // Specify HTTP method
    method: "POST",

    // Define headers (telling server we send JSON)
    headers: {
      "Content-Type": "application/json"
    },
```

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```
      // Convert input values into JSON and send in body
      body: JSON.stringify({
         username: document.getElementById("username").value, // get username
input
         password: document.getElementById("password").value  // get password
input
      })
   })

   // Convert response to JSON
   .then(res => res.json())

   // Display result message from server
   .then(data => {
    document.getElementById("result").innerText = data.message;
    if (data.status == "Success")
         { window.location.href = "/profile.html" }
 })
}

</script>

</body>
</html>
```

Frontend profile.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Profile</title>
</head>
<body>

<!-- Page title -->
<h2>Profile Page</h2>
```

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Hours
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```
<!-- Button to load profile data -->
<button onclick="getProfile()">Load Profile</button>

<!-- Area to display result -->
<h3 id="result"></h3>

<script>

// Function to request profile data from backend
function getProfile() {

  // Send GET request to /profile endpoint
  fetch("/profile")

  // Convert response to JSON
  .then(res => res.json())

  // Display message (welcome or error)
  .then(data => {
    document.getElementById("result").innerText = data.message;
  });
}

</script>

</body>
</html>
```

Student Task

1. Test Login success, Login failure, Access profile without login
2. Modify login credentials
3. Show username in profile page
4. Add logout endpoint (delete cookie)
5. Add logout button in HTML

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Solved for Instructor version

```
// 4: Logout Endpoint
// Create a GET endpoint at /logout
app.MapGet("/logout", (HttpContext context) =>
{
// Delete the "username" cookie from the browser
context.Response.Cookies.Delete("username", new CookieOptions
{
   Path = "/",
   Secure = false,
   SameSite = SameSiteMode.Lax
});
// Return confirmation message
return Results.Ok("Logged out successfully");
});
```

✅ 1. Test Cases (What should happen)

✓ Login Success

- Username: admin
- Password: 123
- Result:
  Login successful
  → Cookie is stored

❌ Login Failure

- Wrong username/password
- Result:
  Invalid username or password

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❌ Access Profile Without Login

- Open /profile.html without logging in
- Result:
  User not logged in

✅ 2. Modify Login Credentials

Change this in **Program.cs**:

```
if (request.Username == "admin" && request.Password == "123")
```

👉 Example:

```
if (request.Username == "student" && request.Password == "456")
```

✅ 3. Show Username in Profile Page (Frontend Improvement)

✓ Updated profile.html

```
<script>
function getProfile() {

  fetch("/profile", {

    credentials: "include"

  })

    .then(res => res.json())

    .then(data => {

      document.getElementById("result").innerText = data.message;

      if (data.status == "Success") {window.location.href="/logout.html" }
```

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```
        })
        .catch(err => {
            document.getElementById("result").innerText = "Error: " + err;
        });
    }
}
</script>
```

👉 Already works, because backend sends:

```
Message = $"Welcome {username}"
```

✅ 4. Add Logout Endpoint (Backend)

✓ Add this to Program.cs

```
// Logout Endpoint
app.MapGet("/logout", (HttpContext context) =>
{

    context.Response.Cookies.Delete("username", new CookieOptions
    {
        Path = "/",
        Secure = false,
        SameSite = SameSiteMode.Lax
    });
```

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```
    return Results.Ok(new
    {
        Status = "Success",
        Message = "Logged out successfully"
    });
});
```

✅ 5. Crea Logout.HTML

```
<!DOCTYPE html>
<html>
<head>
    <title>LogOut</title>
</head>
<body>
    <h2>LogOut Page</h2>
    <button onclick="getLogOut()">LogOut</button>
    <h3 id="result"></h3>
    <script>
        function getLogOut() {
            fetch("/logout", {
                credentials: "include"
            })
```

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```
            .then(res => res.json())
            .then(data => {
              document.getElementById("result").innerText = data.message;
              setTimeout(() => {
                window.location.href = "profile.html";
              }, 1500);
            })
            .catch(err => {
              document.getElementById("result").innerText = "Error: " + err;
            });
        }
    </script>
</body>
</html>
```

✅ Final Result (System Behavior)

Flow:

1. User logs in → cookie stored
2. User opens profile → sees:
   👉 Welcome username
3. User clicks logout → cookie deleted

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4. Open profile again →
   👉 User not logged in

✅ Bonus (Optional Improvement ⭐)

Redirect after login:

```
.then(data => {
  document.getElementById("result").innerText = data.message;

  if (data.status === "Success") {
    window.location.href = "/profile.html";
  }
});
```